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Customer behavioural content recommendation system using decision tree and genetic algorithm for online shopping websites

Swapnil Deshmukh[a] | Ashwini Shinde[b] | Sagar Shinde[c] ✉| Harika Vanam[d] | Rachna Somkunwar[a] | Nirmal Mungale[e]

[a]Department of Computer Engineering, Dr. D. Y. Patil Institute of Technology, Pimpri, Pune, India.
[b]Department of Computer Engineering, Zeal College of Engineering and Research, Pune, Maharashtra.
[c]PCET's - NMVPM's Nutan College of Engineering and Research, Talegaon Dabhade, Pune, India.
[d]Computer Science and Engineering Department, Gokaraju Lailavathi Womens Engineering College (GLWEC), Hyderabad.
[e]G. H. Raisoni College Of Engineering, Nagpur, Maharashtra, India.

Abstract E-commerce has been overgrown due to the ongoing advancements in Internet Technology and the notable enhancement of network fundamentals. Nowadays, the most popular type of electronic commerce is online shopping. These days, user profile-based online recommender systems are popular in engineering and research fields. Getting users' profiles right is essential for making recommendations. Many studies on extracting user profiles, including content-based recommendations, have been initiated recently. We have introduced an intelligent recommendation system for e-commerce, aiming to streamline online shopping by addressing information overload. This system leverages customer behaviour to tailor personalized suggestions and enhance user experiences. It combines a genetic algorithm and a decision tree model for e-commerce platforms. The goal is to smooth the shopping journey, ensuring users find what they seek amidst the vast online options. This approach prioritizes user preferences, reducing the risk of overlooking crucial details when navigating online stores. Through experiments and analysis, we assess its effectiveness compared to conventional recommendation systems. We tested the model using publicly available datasets, comparing its performance against existing recommendation systems. The suggested approach is targeted to decrease recommendation sparsity and increase recommendation accuracy.

Keywords: decision tree, genetic algorithm, online shopping, recommendation system, e-commerce

1. Introduction

In recent years, internet technology has been constantly innovated and developed. As a symbol of digital technology, it impacts every aspect of society and the economy and is now a significant factor pushing upstream consumption. Mobile Internet users are becoming more numerous, and mobile payment systems are developing quickly. Therefore, e-commerce transaction apps are being updated and incorporated into people's lives more (Xu & Sang, 2022; Yuan et al., 2021; Gefen, 2000). Online shopping has emerged as a significant trend in economic development due to the collaboration and integration of offline and online transactions and the effective growth of the logistics express sector (Zha et al., 2006; Stitini et al., 2022; Soares & Viana, 2015).

Businesses may save a lot of money and increase their consumer bases and companies more quickly and easily by using e-commerce platforms for product promotion and sales. The government supports retail reform, leading to businesses combining online and offline approaches to sell products simultaneously. This enhances user experience, adding more value for customers and improving overall satisfaction with their shopping experience. Online shopping allows customers to shop without regard to time or location constraints and facilitates horizontal comparison buying between product kinds and quality. Additionally, consumers may purchase items rarely found in real locations online. In this way, the rapidly expanding field of cross-border e-commerce has met consumer demands.

Massive data mining is the foundation of the personalized recommendation system (Reddy et al., 2019; Oumaima et al., 2021; Verma & Virk, 2015), a sophisticated business intelligence platform. It fulfils consumers' demands and makes product recommendations. An e-commerce recommendation system may significantly increase an online mall's turnover. Amazon has increased the sales volume of an online shopping center by 35% by using a tailored suggestion system. In contrast to conventional search engines, customized recommendation systems employ user behaviour to identify users' interest points

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and help them locate items more quickly. In addition to increasing users' buying efficiency, a robust recommendation system may foster positive relationships with users and increase their feeling of community.

However, problems like data evacuation, low scalability, and difficulties extracting multimedia information features continue to plague conventional recommendation algorithms. Scholars have introduced users' demographic, social, and trust data into similar calculations to address the earlier concerns. These data sets have particular guidelines for addressing data sparsity problems (Omar et al., 2018; Beasley et al., 1993). More studies are needed to handle these issues simultaneously successfully. As a result, a content-based recommendation model for e-commerce online purchasing platforms based on integrating numerous personalized recommendation algorithms, such as genetic algorithms and decision Trees, is put forth. User activity data will be processed using the genuine online shopping platform data set. The platform employs visual analysis to identify pertinent features based on the dimensions of commodities, users, and interactions within them to understand the underlying business logic.

In Figure 1, different recommendation systems are depicted. These include collaborative filtering, content-based filtering, hybrid filtering, and hyper-personalization filtering.

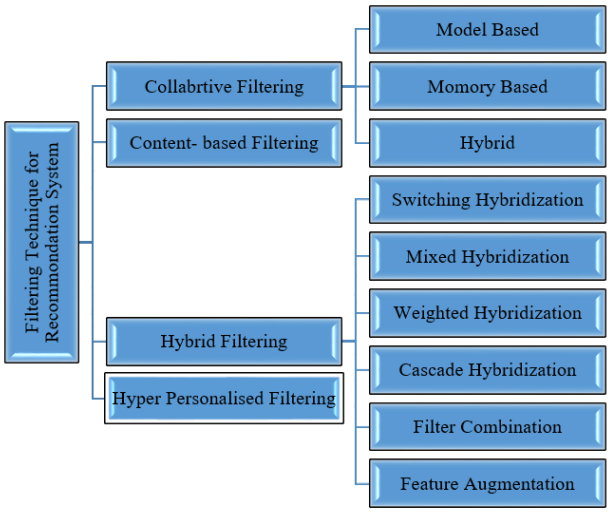

Figure 1 Types of recommendation system.

The following sections comprise the remainder of this article: Section 2 displays critical related research for review. The suggested methodology is described in Section 3. Section 4 presents the outcomes and analysis of the recommended methodology. Section 5 explains the conclusion and future scope.

2. Review

This work tackles the problem of over-specialization in content-based algorithms by focusing on a recommendation system based on evolutionary algorithms (Golbeck, 2006). The main aim of this literature review is to summarize previous studies in relevant areas. Recommender systems are crucial in showcasing unfamiliar yet trending products to customers. These systems utilize various algorithms and methods to generate recommendations. Two popular recommendation methods include collaborative filtering (Verbert et al., 2012), accounting for other users' preferences, and content-based filtering (Hamada et al., 2019), centring on product details instead of user interactions. This review intends to offer insight into the prior research within these domains.

By comparing the properties of different things, the content-based recommendation system algorithm can suggest them (Hwang et al., 2010). For instance, the algorithm would learn to suggest the movie genre that the customer rates highly depending on how they rank other movie genres (Soares & Viana, 2015). A content-based recommendation system uses the user's previous ratings to develop a user profile. Interests are reflected in a user profile, which is flexible enough to accommodate new ones (Reddy et al., 2019). The content-based filtering strategy is based on information acquisition, analysis, and filtering (Mohammadpour et al., 2019). This method is widely applied in handling analyzable data like news, movies, and media carrying metadata. It provides personalized recommendations according to a user's past preferences. Content is labelled for classification, with label weight determined by its accurate description. Leveraging these labels and user preferences, additional content can be recommended to users using nearest-neighbour or clustering algorithms. However, this system faces challenges from novice users lacking experience and limited labels, which can impact its effectiveness and accuracy in generating suggestions.

The predominant approach for content filtering involves analyzing readable or studyable material like news, movies, and media with associated metadata. These recommendations and suggestions are based on a user's past viewing habits. The

weight assigned to each label is based on how accurately it depicts the thing. You may use labels to explain what's within. Depending on the user's preferences and these labels, more articles can be suggested to the current user using nearest-neighbour or clustering algorithms. However, novice users with little information and labels hinder this method. The content-based recommender system has resolved the problems with collaborative systems, yet it has a few shortcomings, such as:

- The amount and type of common traits that content-based approaches may suggest, automatically or manually, is limited. Taxonomies relevant to the particular area, as well as domain knowledge, are also necessary. The content-based recommender system cannot provide relevant recommendations if there is insufficient information in the assessed content.
- An essential technique for locating unexpected information is absent from content-based recommendation systems. Products that fit the user's profile and have a high score are recommended; therefore, the user recommends products similar to those already evaluated. This is sometimes referred to as the over-specialization problem, and it emphasizes how content-based systems typically provide ideas with a low level of originality. A content-based solid strategy is needed to produce some original and unexpected suggestions.
- A content-based recommendation system cannot consider unconventional or unexpected elements in its methodology for suggesting content. Only items with a high score concerning the user profile can be recommended by the algorithm. Due to its ability to demonstrate the boundaries of content-based recommendations, it is sometimes called the serendipity issue. There wouldn't be much fresh material available with a "perfect" content-based approach, which would reduce the range of applications it might be applied to.
- Accumulating a substantial amount of ratings is necessary to develop a recommendation system that can learn about user preferences. The lack of historical data means the algorithm cannot provide new users with recommendations they can rely on.

The recommendation system thoroughly analyses how metadata information included in the descriptions of films and television shows might affect content-based recommendation engines. Researchers discovered that performance improved with more metadata attributes utilized in combination. A method that employs movie content data as the system suggests the training dataset. They (Reddy et al., 2019) proposed an approach that takes textual content data and turns it into feature vectors that can maintain linear connections semantically by extracting features from movie material. A method that recommends movies based on their genres is put out in (Oumaima et al., 2021). A user will be given movies with related genres if they give a movie in that genre a good rating. Another reference (Verma & Virk, 2015) focuses on a system that tailors information to users based on their previous movie preferences and viewing patterns. Employing a hybrid technique, this system integrates content, collaborative processes, and demographic data to refine collaborative filtering accuracy with additional content elements, utilizing the genetic algorithm to suggest items to users.

Genetic algorithms find diverse applications in recommendation systems, notably in clustering, hybrid models, and utilizing genetic algorithms without the supplementary information provided by hybrid models. Each clustering strategy in recommendation systems has strengths and limitations, significantly influencing the accuracy of recommendations for users actively seeking suggestions (Hassan & Hamada, 2018). In (Shani & Gunawardana, 2011), a hybrid evolutionary approach merges genetic algorithm and Gravitational Emulation Local Search (GELS) techniques to optimize item clustering within the collaborative filtering recommender system. This method aims to enhance suggestion accuracy by improving data clustering quality. Similarly, (Adomavicius & Kwon, 2012) introduces hybrid recommendation systems employing genetic algorithms and collaborative filtering, amalgamating data from various sources like product, customer, and transaction records. This integration generates comprehensive consumer preference profiles to enhance recommendation accuracy.

These references underscore the versatility of genetic algorithms in recommendation systems, showcasing their utility in optimizing clustering, refining collaborative filtering accuracy, and integrating diverse data sources to offer improved suggestions to users. Each approach demonstrates how genetic algorithms uniquely enhance recommendation systems in distinct ways, catering to the varied needs of users seeking personalized recommendations. Next, a vector of feature weights—used to determine how similar consumers are to one another—is optimized using genetic algorithms.

In (Hamada et al., 2019), a novel genetic algorithms-based recommender system (BLIGA) based on prior rating data and semantic understanding is put out. They introduce a collaborative filtering approach with many filtering levels. The suggested method generates suggestions for currently engaged people using a genetic algorithm approach. The primary goal was to find a list of strongly linked items with two qualities: they are evaluated by AU neighbours and have the potential to be favourites due to their high anticipated semantic rating values.

The shortcomings of the current recommendation systems are depicted in Figure 2, which includes accuracy and time efficiency as a few critical quality attributes that must be focussed. This uniqueness in recommendations is essential in any recommendations uncovered in this research.

Figure 2 Challenges of existing content-based recommendation systems.

3. Objectives

The following objectives were developed in response to the shortcomings of the current content recommendation system and the recognized need for more study. These are the research goals for creating a recommendation system to generate user product recommendations adequately.

1. Designing a recommendation system for online shopping sites involves using a decision tree alongside a genetic algorithm. This system aims to provide personalized suggestions to users based on their preferences and browsing history.
2. The objective is to create a content-based recommendation system that employs genetic algorithm optimization. This approach focuses on generating unique recommendations by analyzing the content and characteristics of products, enhancing the user experience.
3. The aim is to construct a content-based recommendation system for online shopping platforms that delivers remarkably accurate suggestions to users. This system will utilize various data points to understand user preferences and behaviour, offering tailored recommendations for an improved shopping journey.

4. Proposed Methodology

Our initial effort was a simple recommendation engine that enumerates all comparable items and products according to the user's preferences. Using content-based filtering with a decision tree algorithm makes the primary suggestion. For instance, if the user has already searched for some electronic product within some price range, the primary content-based suggestion will suggest all similar products in the user's genre.

As seen in Figure 3, the suggested approach architecture functions as follows: Before being sent to the decision tree classifier, the browsing attribute values are used to classify the user's search type. The user is classified using a decision tree classifier based on their browsing history, and following classification, recommendations for content-based filtering are produced. A recursive procedure is used to generate decision trees. A training or input set is utilized at the root node to start the process. At each stage of the process, a specific characteristic of an item is chosen as the dividing factor. When a parent node is divided among its child nodes, each child node receives the items that match the chosen characteristic. A new child node is created for every possible value or group of values of this characteristic. The selection of this dividing characteristic is not deterministic. Predicting which division will create the best tree that yields the most favourable outcomes for future inputs is challenging. For instance, the renowned C4.5 method uses a heuristic approach to pick the division that generates the most information gain among all possible divisions. One of the predefined characteristics is the target attribute. The recursive process continues until all items in a node share the same value for the target attribute or until the total number of items exceeds a set limit. Each leaf node is given a label to help categorize its collection of items. This labelling process repeats until all nodes have been appropriately categorized.

The most common value or the shared target attribute value is applied to each leaf node if there are many such values. Ratings comprise the input set used in recommender systems to construct decision trees. A relation Product_ID, User_ID, Rating can be used to represent ratings, where Product_ID, User_ID is thought to be the primary key. Certain factors, such as

age, gender, and employment, might be used to characterize the consumers. Some attributes that may characterize an item include its weight, price, and size. The decision tree uses rating as the goal characteristic for classification. Using data from the training set, the system tries to estimate the items a user has not yet rated. Then, it suggests items to the user based on the ones predicted to have the highest ratings. We used a cosine similarity function to construct a similarity matrix between the products and identify the most comparable items. We looked through the matrix table for every product with a score of more than 0.75. This similarity-based recommendation list is now applied to the genetic algorithm to optimize the same and generate the most accurate recommendation results.

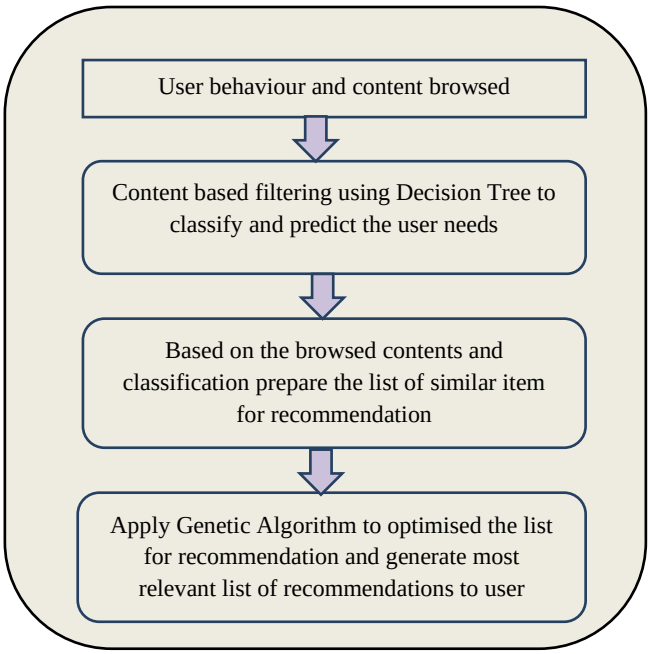

Figure 3 Architecture of recommendation system.

The genetic algorithm is applied as follows:

1. Initial Population:

Our product catalogue encompasses various genres across different products. To streamline our suggestions, we aim to condense our options into five critical choices, each represented by a specific product ID. These IDs will serve as unique markers for individual products. The initial step in our genetic algorithm involves establishing a sample population. This population comprises N randomly selected solutions from the overall pool of possibilities.

Since each chromosome contains five product IDs, the selection process is randomized. Consequently, the algorithm starts searching for optimal solutions based on actual data. Our primary population needs to mirror existing market offerings, as this will help us identify similar options for comparison. Generating this initial population involves a random selection process, resulting in a population size of 290. Each individual within this population will choose from the five available products within our system. This approach to population selection aligns with our overarching goal: identifying individuals who share similar tastes with the user while fostering unexpected discoveries. We intend to pinpoint the most suitable matches by tapping into the uniqueness of each person's preferences.

For clarity, we used a small group of individuals to highlight our initial population. Hence, the table contains only five individuals, each represented according to Equation (1), which outlines the various movie formats available within our system. This format serves as a template for depicting individual preferences. Equation (1) remains our guide for showcasing diverse potential movie formats, offering a glimpse into the range of choices available in our system.

$$[1, 2, 3, 4, 5] \qquad (1)$$

2. Fitness function:

We suppose that the user searched for a product with "electronics" as the first genre and "sound" as the final in our example.

$$[12, 116, 26, 39, 106, 113, 124] \qquad (2)$$

About the first person given in Equation (2): The first product in this person is called 12, and it belongs to the electronics, sound, and Bluetooth genres. To establish a score of 10, they both favored electronics as the first category and sound as the second. Dataset taken from Kaggle https://www.kaggle.com/datasets/lakshmi25npathi/online-retail-dataset. This dataset comprises transactions that took place for an online retail store. The business primarily offers distinctive gifts suitable for any occasion. A significant portion of the company's clientele consists of wholesalers.

Algorithmic steps: Implemented using Python as a Software Tool.

- Data Preparation: Clean and prepare the data for analysis.
- Feature Extraction: Extract relevant features from the user's browsing history.
- Classification: Use a decision tree classifier to classify users based on their browsing history.
- Recommendation Generation: Generate personalized recommendations for users.
- Recursive Tree Generation: Recursively builds a decision tree to optimize user classification and recommendation generation. We have developed using Python with a proposed basic framework to understand how the approach works, which can be expanded and customized based on specific dataset requirements and application needs.

5. Results and Discussion

Part of assessing feedback systems involves addressing an issue and applying an assessment approach to see how the problem has been handled. Several goods and actual consumer search keywords from the Home Depot website are included in this data collection. Predicting a relevancy score for the given sets of items and search phrases is a problem. A value ranging from 1 (not relevant) to 3 (very relevant) indicates the significance. For example, a search for "AA battery" might return three results: very relevant for a pack of size AA batteries, slightly relevant for a cordless drill battery, and irrelevant (relevance = 1) for a snow shovel.

This section includes a discussion of all experiment findings. The gathered data is also examined to contrast the methodologies used in those trials. A sensitivity analysis was carried out to identify the appropriate parameters that result in the optimum genetic algorithm performance. The chosen parameter values are shown in Figure 4.

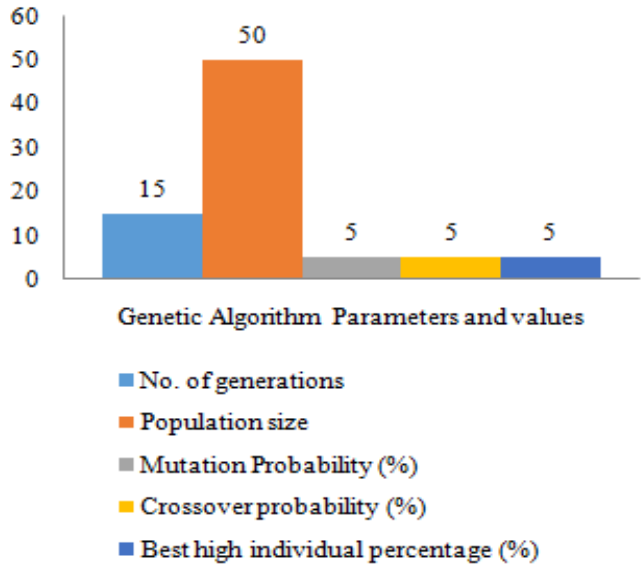

Figure 4 Parameters employed in the genetic algorithm experiment.

Table 1 shows a comparative analysis of our proposed method against alternative recommender system approaches. The genetic algorithm, which produces fortuitous and unexpected items in recommendation lists, produces a higher diversity of results than traditional content-based filtering. Moreover, our proposed method lessens the impact of over-specialization.

Table 1 A comparison between our suggested method and the existing recommender system method.

Properties	Content-Based Filtering	Clustering	Proposed Approach [Decision Tree-GA]
Variety in outcomes	Less	Medium	High
Ability to grow	Low	Low	Good
Impact of being too specialized	High	Medium	Less

The collected uniqueness findings are displayed in Figure 5. One essential aspect of a recommendation system is its uniqueness, significantly enhancing its effectiveness. Adding new items to the list of suggestions improves accuracy and impacts how unfamiliar these recommended items are to the consumer. Uniqueness in a recommendation system gauges its ability to introduce users to content that's both novel and unforeseen, content they're less likely to already know about. This uniqueness factor is quantified by assessing the self-information of recommended items, which helps create an average self-information for all users based on the top N recommendations. We can determine how successfully the system introduces users to new items by analyzing feedback. Our objective is to offer consumers new and unexpected content, and we have devised a simple metric for evaluating this called the discovery index. This index is calculated by dividing the total number of suggestions by the count of items the user isn't aware of (as represented in Equation (3)). Uniqueness plays a vital role in recommendation systems by ensuring users are exposed to fresh and unexplored content, enriching their experience and enhancing the system's accuracy.

$$Uniqueness = \frac{(Number\ of\ recommended\ items\ unknown)}{Total\ number\ of\ provided\ recommendations} \qquad (3)$$

Table 2 shows that the uniqueness values of the proposed approach are better than those of the traditional content-based filtering technique.

Table 2 Uniqueness results of the recommendation methods.

Method Recommendation	Content-based filtering	Our Model
K=1	0.285	0.790
K=4	0.289	0.715
K=6	0.297	0.698
K=8	0.316	0.610
K=10	0.410	0.595

Figure 5 shows the performance comparison of the proposed model architecture using the genetic algorithm with the previous attempts by researchers for recommendation systems using various soft computing techniques. The system's performance is compared using the uniqueness factor for these systems. The genetic algorithm has given an average uniqueness value of **0.629**, superior to the existing approaches.

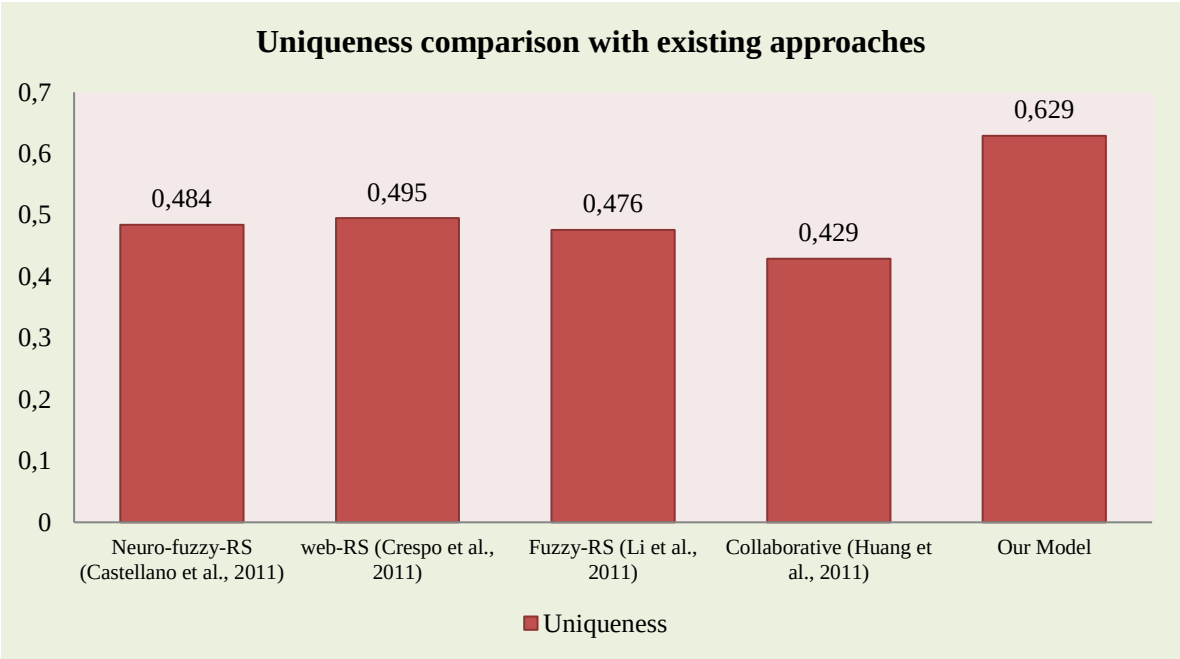

Figure 5 Uniqueness comparison of the content-based recommendation system of our model with existing recommender system models.

The Customer Behavioural Content Recommendation System employs a Decision Tree and Genetic Algorithm to enhance the performance of online shopping websites. In addition to this, we can incorporate supplementary evaluation metrics like user satisfaction and real-time performance. Figure 6 below illustrates the User Satisfaction and Real-Time Performance of various recommendation models.

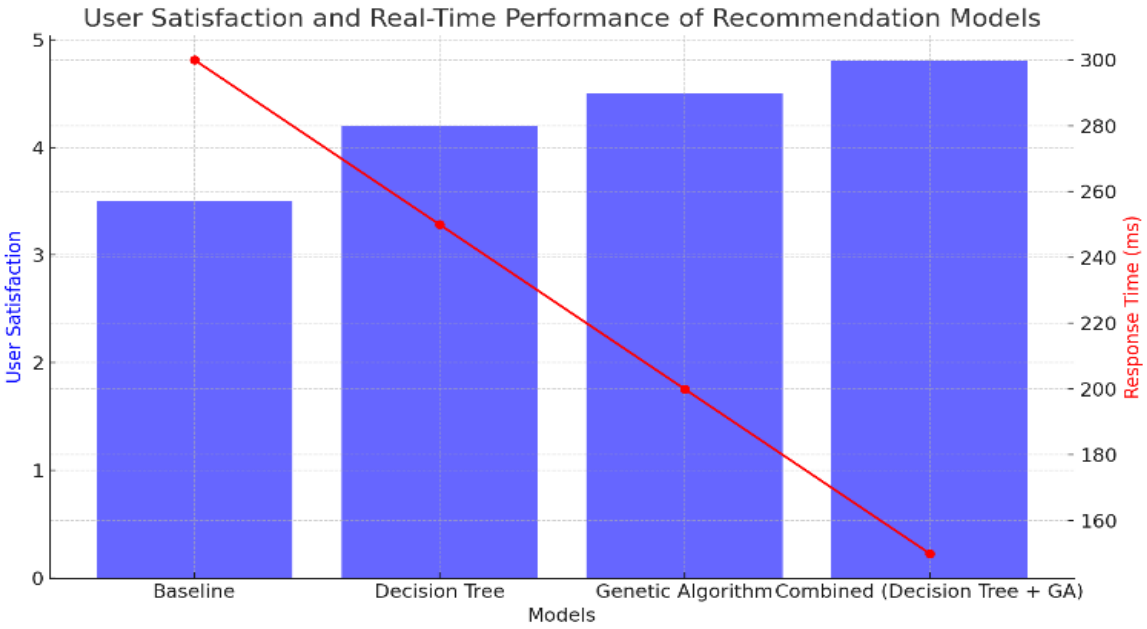

Figure 6 User satisfaction and real-time performance of the recommended model.

6. Conclusion

Recommender systems are becoming more and more critical due to information overload. We aimed to generate innovative concepts for users and explore a new approach to tackle the problem of over-specialization in content-based recommendation systems. We conducted a content-based filtering model, employing a genetic algorithm to offer user suggestions. The main objective of this system is to identify a collection of unique items highly likely to match user preferences based on a robust correlation (the proposed fitness function). Rather than looking for the most remarkable things to construct the list, our model seeks the proper recommendation.

Consequently, the starting population comprises a set of people representing a random set of potential recommendation lists. Our model uniquely tackles the issue of over-specialization by viewing every member of the population as a possible list of recommendations. As a result, the recommended method does not create the suggestion list using a similarity measure. Instead, it tests a potential suggestion list using the similarity metric. We experimented with the model using the Home Depot dataset to tackle the challenge of restricted content through content-based filtering. We compared different approaches with our recommended system regarding accuracy and efficiency in making recommendations. Our findings indicate that the model outperforms other methods in predictive accuracy.

There's potential to expand the system beyond its current small-scale implementation by increasing the dataset size for training and testing. Moreover, there's room for further improvements and enhancements to refine its capabilities in the future. Employing hybrid approaches that combine various recommendation techniques can refine suggestions based on user preferences. Enhancing efficiency requires optimizing algorithms and employing parallel processing for faster computations, allowing real-time recommendations. Additionally, integrating user feedback loops ensures continual system refinement. Incorporating newer data sources and applying novel evaluation metrics could further gauge recommendation quality. Experimenting with novel AI architectures or leveraging natural language processing for improved contextual understanding could augment recommendation system's capabilities.

Ethical considerations

Not applicable.

Conflict of Interest

The authors declare no conflicts of interest.

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